

Uveal Melanoma in Teenagers and Children

A Report of 40 Cases

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Abstract: A review of 3706 consecutive patients with uveal melanoma revealed that 40 patients (1.1%) were age 20 years or younger at the time of diagnosis. The youngest patient was age 6 years but the majority of patients (78%) were between 15 and 20 years old. The tumor occurred in the iris in 5 cases (12%) and in the posterior uvea in 35 cases (88%). The mean largest tumor dimension and thickness was 10 mm and 5 mm, respectively. In all cases, the diagnosis of uveal melanoma was suspected before referral, and misdirected treatment was avoided. The tumor was initially treated by enucleation in 24 cases (60%), local resection in 7 (18%), plaque radiotherapy in 3 (8%), and observation in 6 (15%). Secondary treatment was required in 7 cases in the form of enucleation (4 cases), ablative laser (1 case), plaque radiotherapy (1 case), and exenteration (1 case). The mean follow-up period was 68 months (median, 48 months) from the time of treatment, and only one patient died of metastases (from a massive ciliochoroidal melanoma 33 months after treatment). The remainder of the group of young patients are alive and healthy. Cumulative survival rates show that 96% of young patients with uveal melanoma survive at the 5-year period. *Ophthalmology* 1991; 98:1662-1666

Uveal melanoma occurs at an average age of 55 years and is rare in children and teenagers.¹ Various reports have documented the occurrence of uveal melanoma in children, infants, and even newborns.²⁻¹³ It has been found that 0.6% to 1.6% of all uveal melanomas in a large pathology laboratory were from children and teenagers younger than 20 years of age.^{14,15}

Several overlapping studies on uveal melanoma in the preadult have originated from the Armed Forces Institute of Pathology (AFIP) and have suggested that the tumor in children and adolescents does not differ significantly in clinical appearance or prognosis from its adult counterpart.¹⁴⁻¹⁶ In contrast, other studies have indicated that metastatic disease from childhood uveal melanoma is extremely rare, suggesting that prognosis is more favorable in children than in adults.¹⁷⁻¹⁹

In past reports, the collected series of young patients with uveal melanoma have originated mostly from enucleated globes or specimens from pathology laboratory files and pathology club cases, some of which date back to the days before the use of indirect ophthalmoscopy and more accurate diagnoses.¹⁴⁻¹⁶ In one series of young patients with uveal melanoma, misdiagnoses with misdirected surgical procedures were performed in 9 of 78 cases (12%) and 5 of these 9 patients died of metastases.¹⁵ In this report, we review our more recent clinical experience over the past 17 years in personally managing 40 children/teenagers with uveal melanoma.

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